

Fibrinogen Activity

Test Details

METHODOLOGY

The measurement of fibrinogen activity is based on determination of fibrin polymerization function by the Clauss method.¹⁰ This method measures the rate of clot formation after adding a high concentration of thrombin to citrated plasma. The fibrinogen activity is then derived from a standard curve relating the clotting time to plasma standards of known fibrinogen activity.

RESULT TURNAROUND TIME

Within 1 day

Turnaround time is defined as the usual number of days from the date of pickup of a specimen for testing to when the result is released to the ordering provider. In some cases, additional time should be allowed for additional confirmatory or additional reflex tests. Testing schedules may vary.

Use

Diagnosis of homozygous and heterozygous fibrinogen deficiency as well as dysfibrinogenemia; diagnosis of disseminated intravascular coagulation;⁶⁻⁸ fibrinogen levels can be used to assess the effectiveness of thrombolytic therapy.⁹

Special Instructions

If the patient's hematocrit exceeds 55%, the volume of citrate in the collection tube must be adjusted. Refer to **Coagulation Collection Procedures** (/node/455) for directions.

Limitations

Fibrinogen is an acute-phase reactant and can often become significantly increased in conditions involving tissue damage, infection, or inflammation.⁶ Increased levels may be seen in smokers, during pregnancy, and in women taking oral contraceptives.⁶ Fibrinogen levels can be diminished in advanced liver disease.⁹ Very high levels of heparin, presence of direct thrombin inhibitors, or fibrin breakdown products may falsely reduce fibrinogen levels because they interfere with the rate of clot formation.⁶ Lipemia or hemolysis may interfere with this assay.

Custom Additional Information

Fibrinogen, also referred to as factor I, is a 340-kilodalton glycoprotein that is produced by the liver.⁶ Fibrinogen has a plasma half-life of about four days. Proteolytic conversion of fibrinogen to fibrin occurs through both the extrinsic and intrinsic pathways.⁶ Severe fibrinogen deficiency should be considered when a patient with bleeding history has both extended protime (PT) and activated partial thromboplastin time (aPTT).^{7,8} Mild deficiency may not produce prolongation of either the aPTT or PT and, therefore, fibrinogen activity should be measured in individuals with a bleeding tendency even when the aPTT and PT are in the normal reference interval.

Congenital afibrinogenemia, a condition associated with the complete absence of fibrinogen, is rare with only about 150 cases reported in the literature.^{6,7} Fibrinogen deficiency is inherited as an autosomal recessive trait.^{7,8} Afibrinogenemia occurs in individuals who are homozygous or doubly heterozygous for mutations. These individuals have infinite protime and aPTT results due to the inability to produce fibrin. Approximately 25% of patients with afibrinogenemia have mild thrombocytopenia.⁷

Individuals who are heterozygous for congenital fibrinogen deficiency are usually asymptomatic unless their fibrinogen levels fall to <50 mg/dL.⁷ Both functional (activity) and antigenic levels are diminished in these individuals.⁷ Fibrinogen deficiency affects both males and females with a prevalence that is equal in all ethnic groups.⁷ Acquired deficiencies occur in individuals with significant hepatic dysfunction, renal disease, and after L-asparaginase therapy.⁶ Diminished levels can also be seen in patients with disseminated intravascular coagulation (DIC) or who are undergoing thrombolytic therapy.⁶ Fibrinogen is one of the major determinants of the erythrocyte sedimentation rate and individuals with afibrinogenemia typically have greatly extended sedimentation rates.⁷

Individuals with dysfibrinogenemia have fibrinogen that is qualitatively defective with low functional fibrinogen levels (activity) and normal or decreased antigenic levels.⁶ Congenital dysfibrinogenemia is inherited as an autosomal dominant mutation.⁶ A number of dysfibrinogenemic defects have been identified with a variety of manifestations including abnormal fibrin polymerization, impaired fibrinopeptide release, abnormal fibrin stabilization, and abnormal fibrin clot lysis.^{6,7} Fibrinogen activity and antigen levels are useful in the diagnosis of dysfibrinogenemia since these individuals often have diminished activity relative to antigen levels.⁸ Typically, dysfibrinogenemia is associated with an elevated thrombin time and greatly elevated reptilase time.

Individuals with afibrinogenemia have a bleeding tendency of varying severity.⁷ Symptoms often start in early infancy with umbilical cord bleeding, intracerebral hemorrhage, or bleeding at circumcision.⁶⁻⁸ Individuals with afibrinogenemia also suffer from deep muscle and joint bleeding and other mucous membrane bleeding throughout life.⁶ Women with afibrinogenemia typically do not experience menorrhagia.⁸ Patients with heterozygous hypofibrinogenemia usually have a minimal history of bleeding with symptoms only observed after major surgery or trauma.^{6,7} Approximately 50% of individuals with dysfibrinogenemia are asymptomatic suffering neither bleeding nor thrombosis.^{6,7} These individuals are usually detected when prolonged clotting times are discovered as a result of routine laboratory testing. About one in four will suffer prolonged bleeding after surgery and approximately 20% will have an increased tendency toward thrombosis.⁶

A number of clinical and epidemiological studies have revealed a consistent association between elevated fibrinogen levels and increased risk for atherosclerotic vascular disease;¹¹ however, it remains to be determined whether increased fibrinogen acts as a mediator of arterial thrombosis or simply reflects the inflammation associated with atherosclerosis.¹¹

Specimen Requirements

Specimen

Whole blood **or** plasma

Volume

4.5 mL, 2.7 mL, 1.8 mL

Minimum Volume

90% of full draw (**Note:** This volume does **not** allow for repeat testing.)

Container

Blue-top (sodium citrate) tube; do **not** open tube.

Collection Instructions

Blood should be collected in a blue-top tube containing 3.2% buffered sodium citrate.¹ Evacuated collection tubes must be filled to completion to ensure a proper blood to anticoagulant ratio.^{2,3} The sample should be mixed immediately by gentle inversion at least six times to ensure adequate mixing of the anticoagulant with the blood. A discard tube is not required prior to collection of coagulation samples.^{4,5} When noncitrate tubes are collected for other tests, collect sterile and nonadditive (red-top) tubes prior to citrate (blue-top) tubes. Any tube containing an alternate anticoagulant should be collected after the blue-top tube. Gel-barrier tubes and serum tubes with clot initiators should also be collected after the citrate tubes. If testing cannot be completed within 24 hours, specimens should be centrifuged for at least 10 minutes at 1500xg. Plasma should then be transferred to a Labcorp PP transpak frozen purple tube with screw cap (Labcorp No. 49482). Freeze immediately and maintained frozen until tested. Refer to Coagulation Collection Procedures for directions.

Please print and use the *Volume Guide for Coagulation Testing*

(/tests/related-documents/L5901) **to ensure proper draw volume.**

Stability Requirements

Temperature	Period
Room temperature	1 day
Frozen	Plasma: 14 days

Reference Range

- <6 months: Not established
- 7 months to 16 years: 180–383 mg/dL

• ≥ 17 years: 193–507 mg/dL

Storage Instructions

Specimens are stable at room temperature for 24 hours. If testing cannot be completed within 24 hours, specimens should be centrifuged for at least 10 minutes at 1500xg. Plasma should then be transferred to a LabCorp PP transpak frozen purple tube with screw cap (LabCorp N° 49482). **Freeze** immediately and maintain frozen until tested. Refer to Coagulation Collection Procedures (/node/455) for directions.

Causes for Rejection

Clotted specimen; gross lipemia or hemolysis; tubes <90% full; improper labeling; specimen collected in tube other than 3.2% citrate

References

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Footnotes

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9. Van Cott EM, Laposata M. Coagulation. In Jacobs DS, DeMott WR, Oxley DK, eds. *Laboratory Test Handbook With Key Word Index*. Hudson, Ohio: Lexi-Comp; 2001: 327-358.

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